Experiment 12: Earley Parser for Context-Free Grammar in Python

1. Aim

To write a Python program to implement an Earley parser for context-free grammars.

2. Procedure

1. Import `CFG` and `EarleyChartParser` from the `nltk` library.
2. Define a context-free grammar using `CFG.fromstring()`.
3. Initialize the `EarleyChartParser` with the grammar.
4. Tokenize the input sentence and parse it using `parser.parse()`.
5. Display the parse tree generated by the Earley parser.

3. Program

```
import nltk
from nltk import CFG

grammar = CFG.fromstring("S -> NP VP\nNP -> Det N | 'John'\nVP -> V NP\nDet -> 'the'\nN -> 'ball'\nV -> 'hit'")

parser = nltk.EarleyChartParser(grammar)
sentence = "John hit the ball".split()

for tree in parser.parse(sentence):
    print("Earley Parse Tree:", tree)
```

4. Output

```
Earley Parse Tree: (S (NP John) (VP (V hit) (NP (Det the) (N ball))))
```

5. Result

Thus, the Python program was successfully executed to implement an Earley parser for context-free grammars using `EarleyChartParser`.